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Problem 1 — The inductor will not arrive in time

A. Calculate the original design.

Initial data

$L := 1.7\ \text{nH}$ $C := 0.3\ \text{pF}$ $F := 5.8\ \text{GHz}$ $Rp := 500\ \text{ohm}$ $Cp := 0.12\ \text{pF}$

Calculate the ideal resonance, including Cp.

$Ct := C + Cp$

$$f_0 := \frac{1}{2 \cdot \pi \cdot \sqrt{L \cdot Ct}} = 5.9562 \cdot 10^{9}\ \text{Hz}$$

Find the inductive and capacitive reactance at 5.8 GHz.

$X_L := 2 \cdot \pi \cdot F \cdot L = 61.9522\ \Omega$ $X_C := \frac{-1}{2 \cdot \pi \cdot F \cdot Ct} = -65.3345\ \Omega$

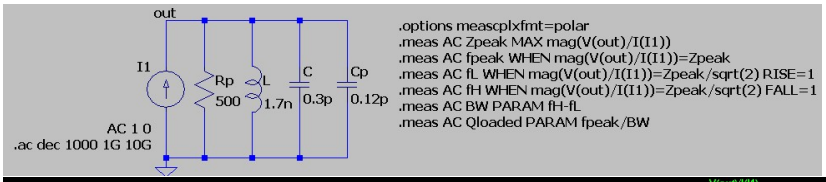
Estimate the loaded Q with Rp = 500 Ω.

$$Q := \frac{Rp}{2 \cdot \pi \cdot f_0 \cdot L} = 7.8591$$

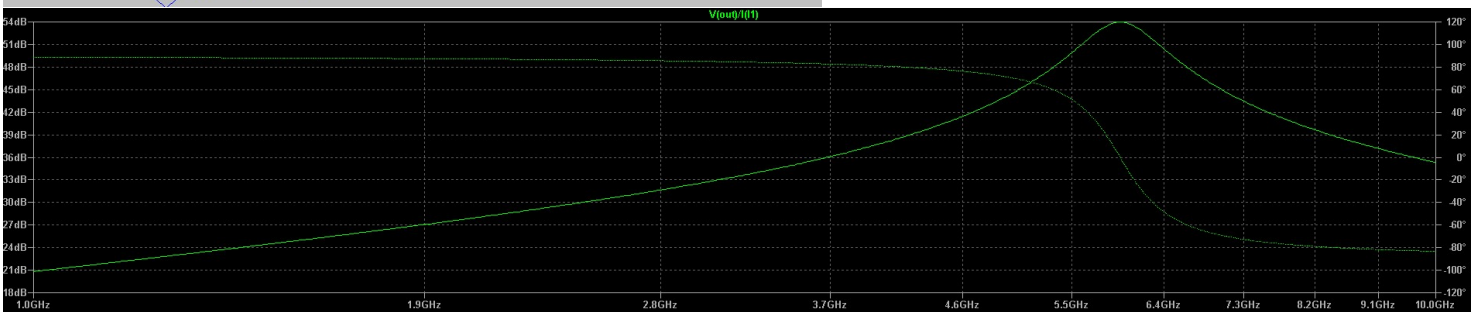
What changes when you include component losses and internal parasitic?

Including component losses and internal parasitics reduces the peak impedance. In the simulated circuit, the resonant peak shifts to a lower frequency and becomes broader. The increased bandwidth, together with the lower peak frequency, results in a lower loaded Q and therefore reduced frequency selectivity.

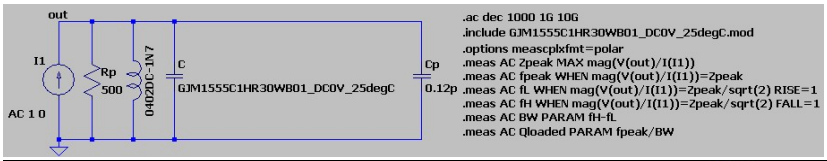
Ideal model and simulation results (Examen Parcial 1 2026\01 original design\simulation.log)



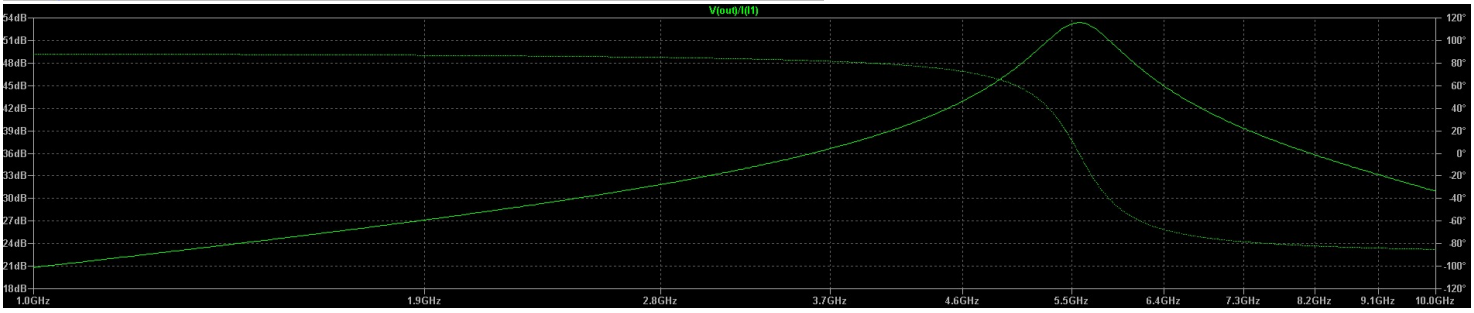

```
zpeak: MAX(mag(V(out)/I(I1)))=(499.999725025,0Â°) FROM 1000000000 TO 10000000000
fpeak: mag(V(out)/I(I1))=Zpeak AT 5956621435.29
fL: mag(V(out)/I(I1))=Zpeak/sqrt(2)  AT 5589329597.18
fh: mag(V(out)/I(I1))=Zpeak/sqrt(2)  AT 6347210355.4
bw: fH-fL=(757880758.226,0Â°)
qloaded: fpeak/BW=(7.85957602253,0Â°)
```



Non-Ideal model and simulation results (Examen Parcial 1 2026\01 original design\simulation.log)




```
zpeak: MAX(mag(V(out)/I(I1)))=(466.705413612,0Â°) FROM 1000000000 TO 10000000000
fpeak: mag(V(out)/I(I1))=Zpeak AT 5571857489.32
fL: mag(V(out)/I(I1))=Zpeak/sqrt(2)  AT 5245291667.14
fh: mag(V(out)/I(I1))=Zpeak/sqrt(2)  AT 5900218137.42
bw: fH-fL=(654926470.276,0Â°)
qloaded: fpeak/BW=(8.50760771202,0Â°)
```



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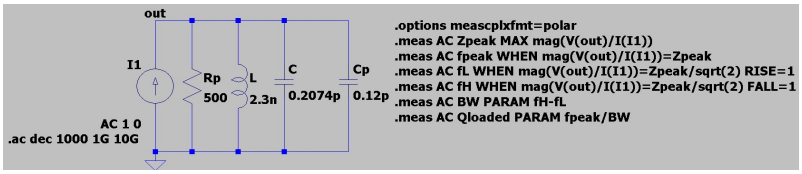
B. Choose a new LC pair.

Choose new L and C values.

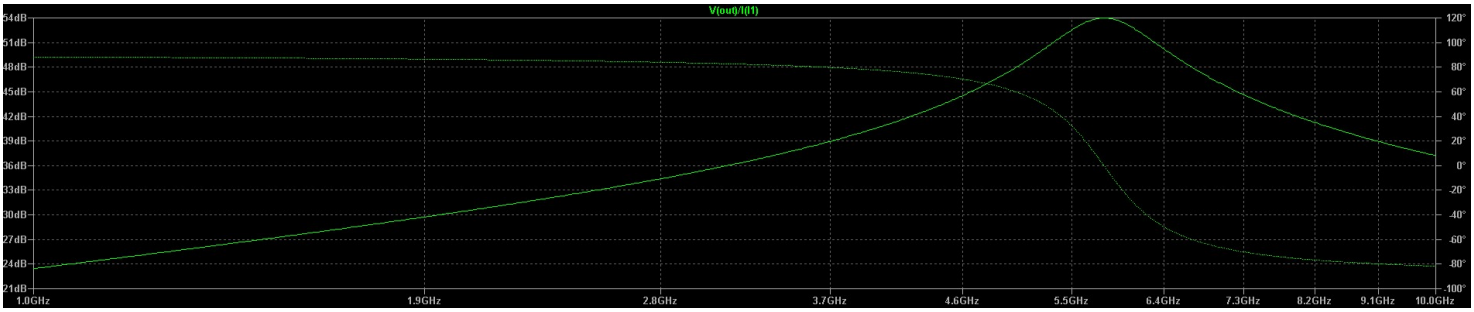
$L := 2.3\ \text{nH}$

$$C := \frac{1}{(2 \cdot \pi \cdot F)^2 \cdot L} - Cp = 0.2074\ \text{pF}$$

The peak impedance of your new tank must be at least 80% of the original. Use the same modelling method for both. Report their loaded Q and bandwidth, and explain the difference. You do not have to obtain the same bandwidth.




```
zpeak: MAX(mag(V(out)/I(I1)))=(499.967213573,0Â°) FROM 1000000000 TO 10000000000
fpeak: mag(V(out)/I(I1))=Zpeak AT 5794286964.27
fl: mag(V(out)/I(I1))=Zpeak/sqrt(2)  AT 5334012830.48
fh: mag(V(out)/I(I1))=Zpeak/sqrt(2)  AT 6306376890.3
bw: fH-fL=(972364059.819,0Â°)
qloaded: fpeak/BW=(5.95896866586,0Â°)
```

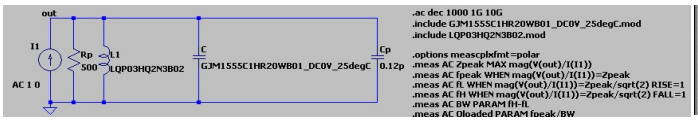


C. Validate commercial components

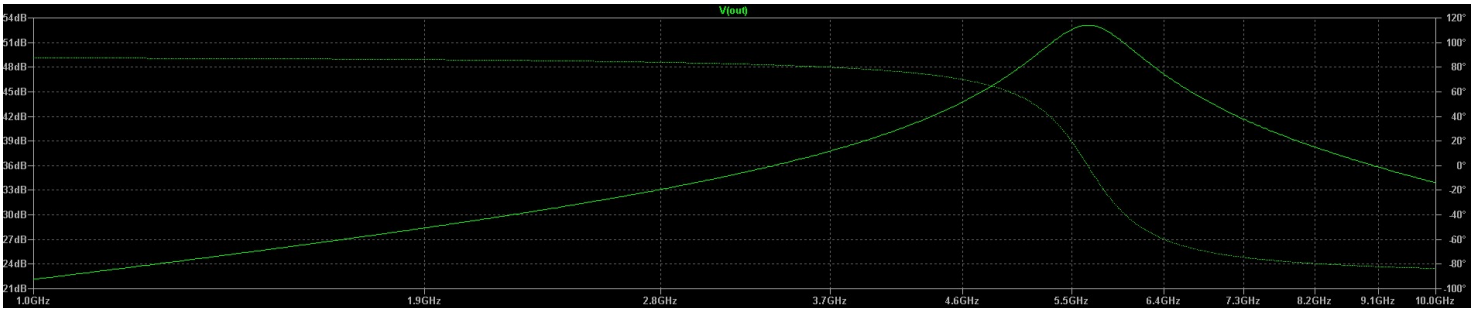
Give the manufacturer and complete orderable part number for each replacement. Use manufacturer data to check tolerances, Q, self-resonance and suitability at 5.8 GHz. State the frequency of each Q value you use.
Include capacitor losses and internal parasitics. State where your model values came from and whether the models cover 5.8 GHz. Explain the difference between component Q and loaded circuit Q.








```
zpeak: MAX(mag(V(out)/I(I1)))=(453.047953409,0Â°) FROM 1000000000 TO 10000000000
fpeak: mag(V(out)/I(I1))=Zpeak AT 5649369748.12
fl: mag(V(out)/I(I1))=Zpeak/sqrt(2)  AT 5244895386.97
fh: mag(V(out)/I(I1))=Zpeak/sqrt(2)  AT 6077473180.57
bw: fH-fL=(832577793.604,0Â°)
qloaded: fpeak/BW=(6.78539566095,0Â°)
```



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D. Additional considerations

For both designs, find the lowest and highest ideal resonance using the L and C tolerances, with Cp held fixed. Show which combinations give these limits.

Initial data

$L := 2.3\ nH \quad L_{tol} := 0.1\ nH \quad C := 0.2\ pF \quad C_{tol} := 0.01\ pF \quad Rp := 500\ ohm \quad Cp := 0.12\ pF$

$Ct_{min} := C + Cp - C_{tol} \quad Ct_{max} := C + Cp - C_{tol} \quad L_{min} := L - L_{tol} \quad L_{max} := L + L_{tol}$

Combination of tolerances to find maximum and minimum values of ideal resonance

$f_0 := \frac{1}{2 \cdot \pi \cdot \sqrt{L_{min} \cdot Ct_{min}}} = 6.0944\ GHz \qquad f_0 := \frac{1}{2 \cdot \pi \cdot \sqrt{L_{min} \cdot Ct_{max}}} = 6.0944\ GHz$ max ideal resonance

$f_0 := \frac{1}{2 \cdot \pi \cdot \sqrt{L_{max} \cdot Ct_{max}}} = 5.8349\ GHz \qquad f_0 := \frac{1}{2 \cdot \pi \cdot \sqrt{L_{max} \cdot Ct_{min}}} = 5.8349\ GHz$ min ideal resonance

E. Availability and cost

State the quotation date and currency. Each circuit uses one inductor and one capacitor. Use the price tier that applies when buying 1,000 of each part.
In two or three sentences, say whether you would order these parts for the prototype. Use your technical results and the availability to justify your answer. Identify any check you still need before ordering. Also state whether the quoted stock is enough for 1,000 circuits.

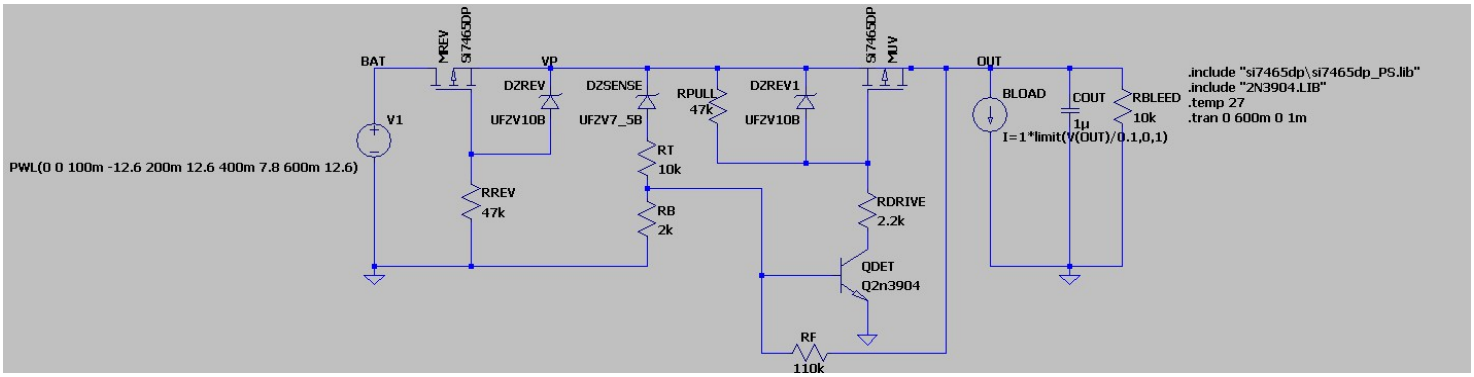
Part Number	Type	Supplier	Stock/lead time	Unit price at 1,000 pieces	Cost for 1,000 circuits
LQP03HQ2N3B02D	Inductor	Murata	133,653	$ 0.07294	$ 72.94000
GJM1555C1HR20WB01D	Capacitor	Murata	161,648	$ 0.04840	$ 48.40000
Total					$ 121.34000

I would order these parts for the prototype after confirming that the new tank's peak impedance is at least 80% of the original design's peak impedance using the same modelling method. The simulated nominal resonance of 5.649 GHz is within the required band. The quoted stock of both components is sufficient for 1,000 circuits.

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Problem 2 — From a 3S battery to 6S



A. How does it work?

Explain how the 3S circuit works.

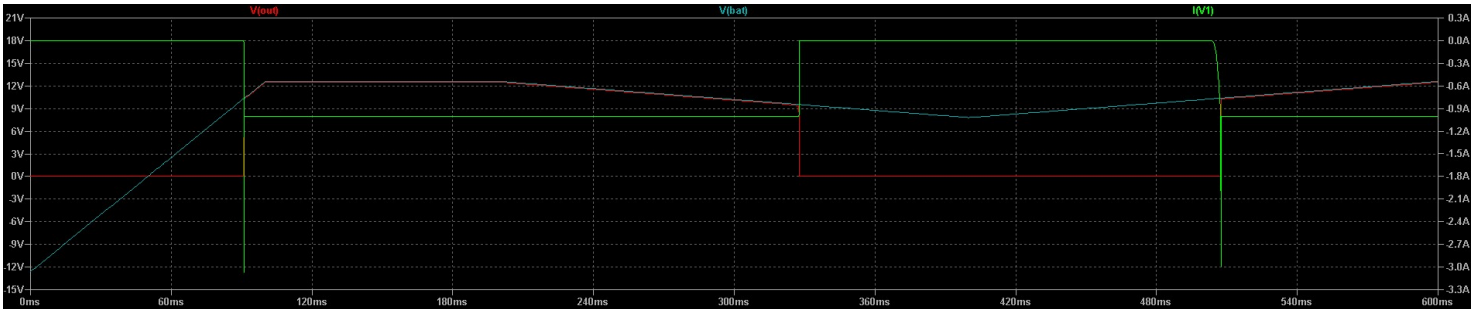
The circuit has 2 main features, the protection against reverse connection and undervoltage.

What do the PMOS devices, QDET and DZSENSE do?

MREV protects the circuit against reverse battery polarity by blocking current when the battery is connected backwards.
MUV disconnects the load when the battery voltage falls below the undervoltage cutoff threshold.
QDET controls MUV by pulling its gate low when the battery voltage is sufficient, turning it on. When the battery voltage is too low, QDET turns off and RPULL pulls MUV's gate toward its source, turning MUV off.
DZSENSE provides a Zener voltage reference that, together with RT, RB, and the feedback resistor RF, sets the cutoff and reconnection thresholds.

Show the current paths during normal operation, after UVLO cuts off, and with the battery reversed. Include the MOSFET body diodes.

In normal operation, the current path is BAT -> MREV channel -> VP -> MUV channel -> OUT -> load -> GND -> BAT.
After UVLO cuts off the path is BAT -> MREV channel -> DZSENSE -> RT -> RB -> GND -> BAT. Also COUT discharges with RBLEED and BLOAD.
In reversed polarity, there is no current path.
This can be observed in below plot.



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B. Adapt it to 6S

Calculate the new values and show your schematic. Which components need to change? Select commercial parts and check their voltage, current and power ratings. Include Zener bias, BJT drive and the gate-source voltage limits at a fully charged battery.

$VZ := 18\ \text{volt}$ $VBE := 0.6\ \text{volt}$ $RT := 10\ \text{kohm}$ $RB := 2\ \text{kohm}$ $RF := 110\ \text{kohm}$

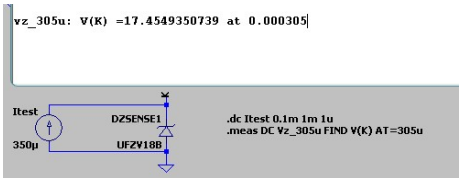
First calculation of VP with 18 V zener diode.

$$VP_{on} := VZ + VBE \cdot \left(1 + \frac{RT}{RB} + \frac{RT}{RF}\right) = 21.6545\ \text{V}$$

$$VP_{off} := \frac{VP_{on}}{1 + \frac{RT}{RF}} = 19.85\ \text{V}$$

Actual VZ at IZ current.

$$Iz := \frac{VP_{on} - VZ - VBE}{RT} = 0.3055\ \text{mA}$$

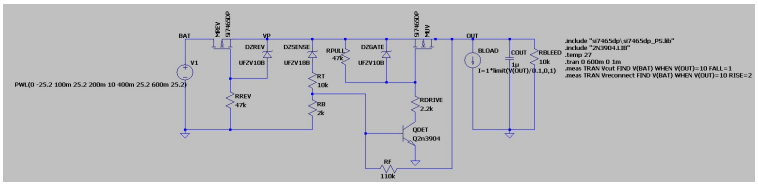


Actual VP on and off.

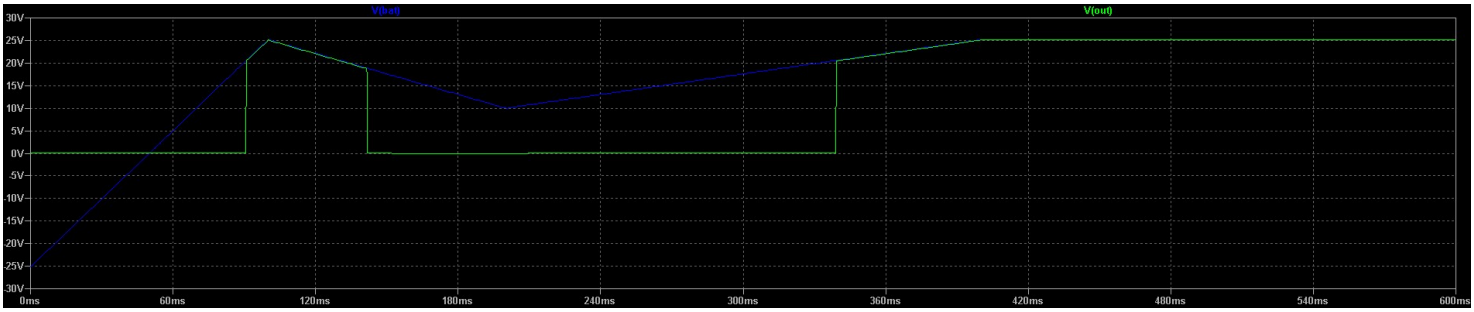
$VZ := 17.45\ \text{volt}$

$$VP_{on} := VZ + VBE \cdot \left(1 + \frac{RT}{RB} + \frac{RT}{RF}\right) = 21.1045\ \text{V}$$

$$VP_{off} := \frac{VP_{on}}{1 + \frac{RT}{RF}} = 19.3458\ \text{V}$$



```
vcut: V(BAT) =18.8628620572 at 0.141691693083
vreconnect: V(BAT) =20.5831149728 at 0.339251502011
```

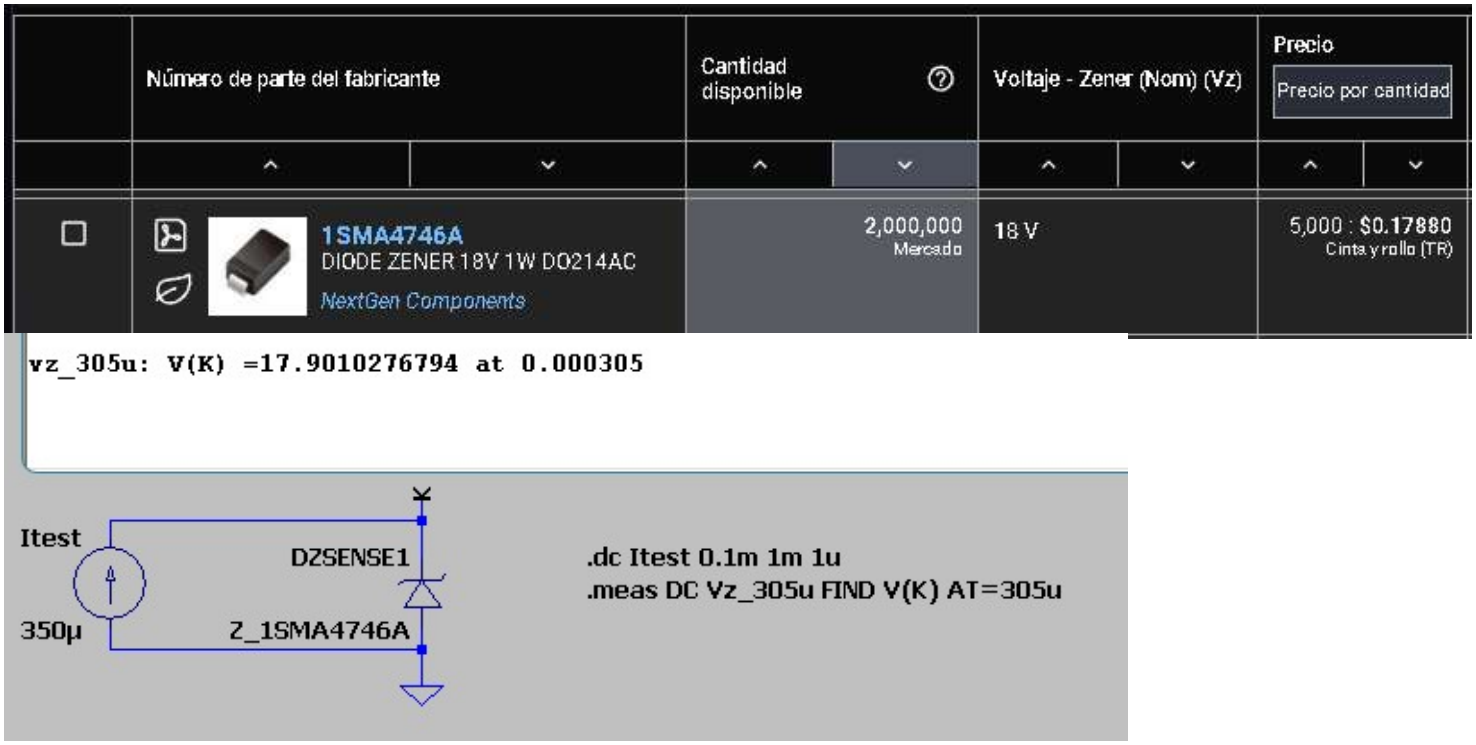


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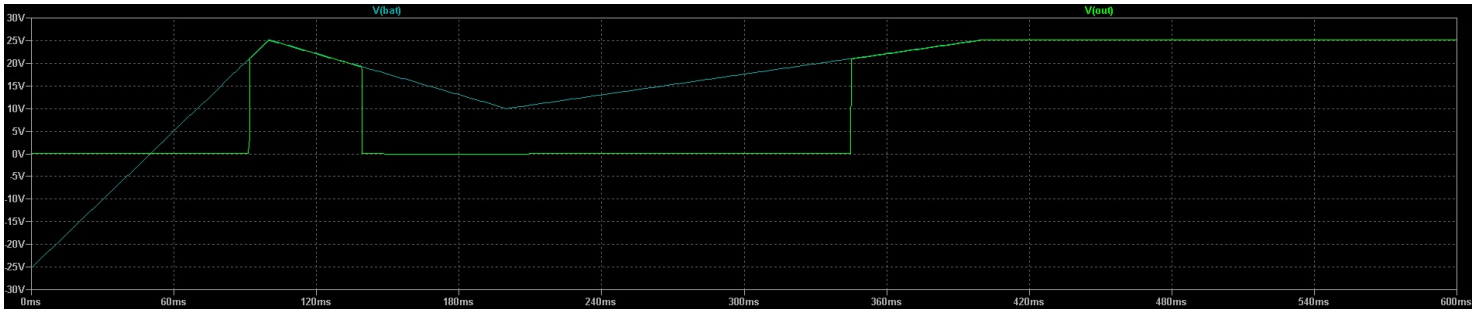
Select commercial parts

In order to select a component which is in stock, the 1SMA4746A was selected, which gave us a better voltage at Iz current



Model simulation with new part number

```
vcut: V(BAT) =19.2561741346 at 0.139104121765
vreconnect: V(BAT) =21.0264475372 at 0.345084827389
```



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C. Diode or MOSFET? Compare both options for reverse-polarity protection

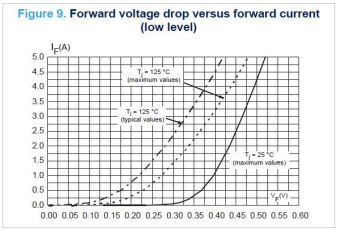
1. Output voltage and losses at 22.2 V input, with 1 A and 2 A loads.

$Vin := 22.2\ V \quad I_{1Amp} := 1\ A \quad I_{2Amp} := 2\ A$

Diode selected STPS5L60S



Diode Output voltage



$Vf_{1Amp} := 0.4\ V$

$Vin - Vf_{1Amp} = 21.8\ V$

$Vf_{2Amp} := 0.44\ V$

$Vin - Vf_{2Amp} = 21.76\ V$

MOSFET output voltage

$Rdson := 64\ mohm$

$V_{M1Amp} := I_{1Amp} \cdot Rdson = 0.064\ V$

$Vin - V_{M1Amp} = 22.136\ V$

$V_{M2Amp} := I_{2Amp} \cdot Rdson = 0.128\ V$

$Vin - V_{M1Amp} = 22.136\ V$

Diode loss

$P_{d1Amp} := Vf_{1Amp} \cdot I_{1Amp} = 0.4\ W$

$P_{d2Amp} := Vf_{2Amp} \cdot I_{2Amp} = 0.88\ W$

MOSFET loss

$P_{m1Amp} := I_{1Amp}^{2} \cdot Rdson = 0.064\ W$

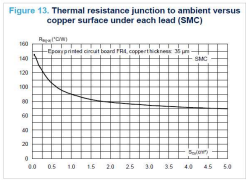
$P_{m2Amp} := I_{2Amp}^{2} \cdot Rdson = 0.256\ W$

A simple temperature estimate at 60 °C ambient and 2 A. State the mounting conditions used for the thermal resistance. Allow for the MOSFET's higher resistance when hot.

$TA := 60\ °C$

Diode

$JA_d := 80\ \frac{K}{W}$



$TJ_d := TA + P_{d2Amp} \cdot JA_d = 130.4\ °C$

MOSFET

$JA_m := 85\ \frac{K}{W}$

$Rdson_{hot} := 1.5 \cdot Rdson = 0.096\ \Omega$

$P_{m2Amphot} := I_{2Amp}^{2} \cdot Rdson_{hot} = 0.384\ W$

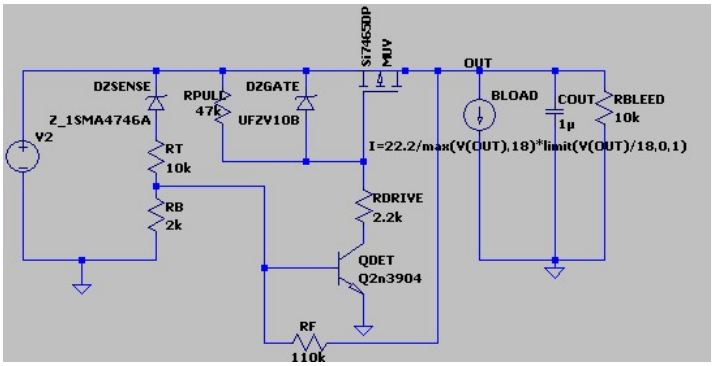
$TJ_M := TA + P_{m2Amphot} \cdot JA_m = 92.64\ °C$

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Relative battery runtime at the same useful load power. For this calculation, use a fixed 22.2 V battery, a 22.2 W external load and equal usable battery energy. Include the control circuit and bleeder losses.

After VP, the rest of the circuit consumes 1.0288277 A (V2 = 22.2 - Diode voltage drop) and 1.0133106 A (V2 = 22.2 - MOSFET voltage drop)



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Problem 3 — Will this LDO run too hot?

A. Power dissipation

Calculate how much power the regulator dissipates.

$Iout := 400\ mA$ $Ignd := 0.70\ mA$ $Vin := 5\ V$ $Vout := 3.3\ V$

$Iin := Iout + Ignd = 400.7\ mA$

$P := Vin \cdot Iin - Vout \cdot Iout = 0.6835\ W$

B. Thermal resistance

Estimate the junction-to-ambient thermal resistance for the available copper area. Indicate which curve you used and include units.

Using the information about PCB (Two-layer FR-4, 1 oz copper) and the area of copper available to spread heat ($100\ mm^2$), the Theta JA can be identified as below.

$JA := 107\ \frac{K}{W}$

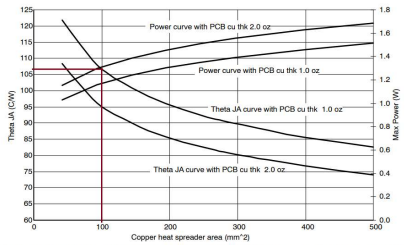


Figure 32. SOT-223 Thermal Resistance and Maximum Power Dissipation vs. P.C.B. Copper Length

Is the regulator's current rating enough to decide whether it is suitable here? Explain briefly.

No. The regulator's current rating alone does not establish that it can continuously supply 400 mA in this design. At that load, it dissipates 0.6835 W, so the junction temperature must be calculated using the PCB's thermal resistance and the 60 °C ambient temperature.

C. Junction temperature

Calculate the junction temperature at the stated load and ambient temperature. Compare it with the operating limit and the required 10 °C margin.

$T_{AMAX} := 60\ °C = 333.15\ K$

$T_j := T_{AMAX} + P \cdot JA = 133.1345\ °C$

Juntion temperature is higher than the maximum operating junction temperatue (125 °C).

Operating Junction Temperature Range	T_{OP}	0 to 125	°C

Explain why the absolute maximum temperature and the thermal shutdown temperature are different limits.

The maximum junction temperature (150°C) is an stress limit. While the thermal shutdown(160°C) is the temperature at which the protection circuit is designed in order to protect the maximum junction temperature is exceeded.

Thermal Considerations

This series contains an internal thermal limiting circuit that is designed to protect the regulator in the event that the maximum junction temperature is exceeded. When activated, typically at 165°C, the regulator output switches off and then back on as the die cools. As a result, if the device is continuously operated in an overheated condition, the output will appear to be oscillating. This feature provides protection from a catastrophic device failure due to accidental overheating. It is not intended to be used as a substitute for proper heatsinking. The maximum device power dissipation can be calculated by:

$$P_D = \frac{T_{J(max)} - T_A}{R_{\theta JA}}$$

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D. Your engineering decision

Would you APPROVE or REJECT this design? Justify either answer with your numbers.

I would reject the design. At a 60 °C ambient temperature, the regulator dissipates 0.6835 W. With the estimated thermal resistance of 107 °C/W for 100 mm² of 1 oz copper, the junction reaches approximately 133.1 °C. This is 8.1 °C above the 125 °C operating limit and 18.1 °C above the 115 °C design target required by the 10 °C margin.

Would increasing the copper area to as much as 500 mm² be enough, keeping 1 oz copper and the same thermal model?
No, to meet the margin at the same load and ambient temperature, both PCB and area would need to be changed, the proposal would be 2 oz copper and a $400\ \text{mm}^2$ area.

$T_{jmax} := 115\ °\text{C} = 388.15\ \text{K}$

$JA := \frac{T_{jmax} - T_{AMAX}}{P} = 80.4682\ \frac{\text{K}}{\text{W}}$

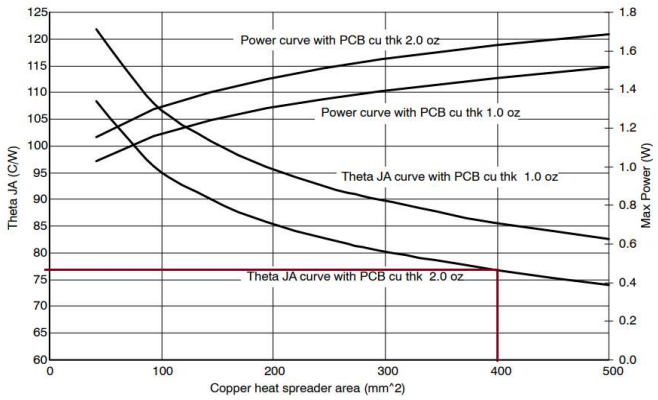


Figure 32. SOT−223 Thermal Resistance and Maximum Power Dissipation vs. P.C.B. Copper Length

Please refer to https://github.com/carloszlopez/Analog-design-class/tree/Examen/Parcial_1_2026 for the solution models

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